MILITARY SPECIFICATION

SIMULATOR, FLASH, ARTILLERY, M21
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of MIL-S-70682 (AR), dated 30 March 1990, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.4.6 Delete in its entirety.

PAGE 5

4.3.1 Change the First Article quantity column of the following parts to read:

Part Description	Drawing	Quantity
Cap, Protecting	9342132	3/cavity
Container, Flash Charge	9342142	3/cavity
Body	9342143	3/cavity
Tube Assembly	9342144	5
Tube	9342145	5
Cover, Flash Charge	9342149	3/cavity
Plug, Electrical Assembly	9342154	3/cavity
Insert, Threaded	9342155	5
Pin, Electrical	9342156	5
Cap, Short Circuit	9342157	3/cavity

Under table, line seven, delete "(less quickmatch)".

Add to table:

Plug, Electrical Assembly (Alternate)	1294400	15
Spring, Short Circuit	12953395	25

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PAGES 6-8

TABLE I: Delete in their entirety and substitute new pages included.

PAGE 9

TABLE 1, Heat Soak: Delete "Heat soak, 108, -, 3.4.4, 4.5.3.4".

Match and Cable Assembly. Under Electrical resistance test, delete "(less quickmatch)".

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TABLE I. Fir article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE: Simulator, Flash, Artillery M21 and Components	SHEET 1 OF 5	DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY: ---
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cap, Protecting (Dwg. 9342132) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.1
	Container, Flash Charge (Dwg. 9342142) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.2
	Tube (Dwg. 9342145) Examination for defects Moisture content	5, 0-1 (a)	3.2 3.3.2	4.4.2.3 4.5.2.1
	Tube Assembly (Dwg. 9342144) Examination for defects	5, 0-1	3.2	4.4.2.5
	Body (Dwg. 9342143) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.6
	Cover, Flash Charge (Dwg. 9342149) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.8

NOTES

(a) Sample for this test shall be taken from test samples.

AMSMC Form 1570, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE: Simulator, Flash, Artillery M21 and Components	SHEET 2 OF 5	DRAWING NUMBER: See Below
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Pin, Electrical (Dwg. 9342156)			
	Examination for defects	5, 0-1	3.2	4.4.2.10
	Insert, Threaded (Dwg. 9342155)			
	Examination for defects	5, 0-1	3.2	4.4.2.11
	Plug, Electrical Assembly (Dwg. 9342154)	3/, 0-1 cavity	3.2	4.4.2.12
	Examination for defects			
	Cap, Short Circuit (Dwg. 9342157)	3/, 0-1 cavity	3.2	4.4.2.14
	Examination for defects			
	Igniter, Electrical Assy (Dwg. 9342148)			
	Examination for defects	15, 0-1	3.2	4.4.2.15
	Electrical resistance test		3.2	4.4.2.15
	Functional timing sequence test		3.4.6	4.5.3.6

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Simulator, Flash, Artillery M21 and Components	SHEET 3 of 5	DRAWING NUMBER See Below
			NEXT HIGHER ASSEMBLY ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Packing and Marking For Box, Packing (Dwg. 9342972) Examination for defects	1 box, 0-1 (162 items)	3.2	4.4.2.24
	Plug, Electrical Assembly (Alternate) (Dwg. 1294400) Examination for defects	15, 0-1	3.2	4.4.2.25
	Spring, Short Circuit (Dwg. 12953395) Examination for defects	25, 0-1	3.2	4.4.2.26

NOTES:

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PAGE 13

4.4.2.2 Delete minor defect 201 in its entirety.

PAGE 15

4.4.2.4: Delete in its entirety.

PAGE 16

4.4.2.5, Major 101: Delete “Gage” and substitute “Visual - straight edge”.

Delete Minor 201 in its entirety.

PAGE 17

4.4.2.6 Delete major defects 101, 102 and minor defects 202, 204 in their entirety,

PAGE 18

4.4,2.7: Case, Cartridge Assembly: Add major defect "101 Pull test, Tube Assembly to Body, 100%, -, 3.2, Gage”.

PAGE 19

4.4.2.8 Change major defect 102 to read:

"102 Slot for cable missing .40% 3.2 Visual”

Delete Major 103 in its entirety.

PAGE 20

4.4.2.9: Quickmatch. Delete in its entirety.

PAGE 21

4.4.2.10 Change major defects 101, 102, and 103 and minor defect 203 to read:

“101 Length from slotted end to flange, near side”.

“102 Length from slotted end to flange, far side”.

“103 O.D. of slotted end before cutting slot”.

“203 Presence of slot .65% 3.2 Visual".

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Add:

"106 Diameter of pin head, -, 0.40%, 3.2, Gage
107 Length of pin head, min, -, 0.40%, 3.2, Gage
108 Width of slots, -, 0.40%, 3.2, Gage"

PAGE 22

4.4.2.11 Add "Note 1" callout following “Major” and "Minor"

Add note:

"1 - These classified defects do not apply to the aternative part which is a commercial item.”

PAGE 23

4.4.2.12 Delete major defects 101 and 104 in their entirety.

Change major defect 102 and 103 to read:

“102 Width under head (Datum B)”.

“103 Length under head (Datum A)".

PAGE 25

4.4.2.14: Cap, Short Circuit. Delete Major 103 in its entirety and substitute “Major 103 Continuity/Functional test, 100%, 3.2, Gage”.

Delete major defect 104 in its entirety.

PAGE 26

4.4.2.15: Major 102, under AQL column, delete “.40%” and substitute "100%. Delete "1/".

Delete Note 1 in its entirety,

Delete major defects 101, 103, 104 and minor defect 201 in their entirety.

Add under major defects:

“When using Plug, Electrical Assembly (ALT), Dwg. 12944400: 105, Match insecure or improperly assembled (soldered) 0.40% 3.2 Visual/Manual"

Add:

"When using Spring Short Circuit, Dwg. 12953395, Major 106, Pull test, spring to pins, 30, 0-1, Gage".

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PAGE 27

4.4.2.16: Delete in its entirety.

PAGE 28

4.4.2.17 Delete major defect 101 in its entirety.

PAGE 31

4.4.2.10 Simulator, Flash, Artillery M21 (prior to assembly of protecting cap): Critical 1: Delete “Short circuit cap missing or damaged” and substitute “Short Circuit device (cap or spring) missing or damaged.”

PAGE 32

4.4.2.21: Delete major 102 in its entirety.

PAGE 36A

4.4.2.25: Add new page 36A included.

PAGE 36B

4.4.2.26: Add new page 36B included.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4 2.25	Plug, Electrical Assembly (ALT)	1 OF 1	12944400
			NEXT HIGHER ASSEMBLY 9342148

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of body, max		0.40%	3.2	Gage
102	Width under head (Datum A)		0.40%	3.2	Gage
103	Depth under head (Datum B)		0.40%	3.2	Gage
104	Pins missing or damaged		0.40%	3.2	Visual
105	True position of pins		0.40%	3.2	Gage
106	Pin insert shoulders above flush, max.		0.40%	3.2	Gage
107	Pull test, pins to plug body	30	0-1	3.2	Gage
108	Pull test, cable to plug body	30	0-1	3.2	Gage
109	Diameter of pin head		0.40%	3.2	Gage
110	Length of pin head, min.		0.40%	3.2	Gage
111	Width of slots		0.40%	3.2	Gage
112	Length of shorting clip, max.		0.40%	3.2	Gage
113	Thickness of shorting clip		0.40%	3.2	Gage
114	Width of shorting clip, min.		0.40%	3.2	Gage
115	Width of shorting clip strap		0.40%	3.2	Gage
Minor					
201	Length of cable		0.65%	3.2	Gage
202	Shorting clip strap improperly assembled		0.65%	3.2	Gage
203	Length of shorting clip strap, max.		0.65%	3.2	Gage
204	Poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Spring, Short Circuit	1 OF 1	12953395
			NEXT HIGHER ASSEMBLY 9342148

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	ID of coils		0.40%	3.2	Gage
102	Centerline distance		0.40%	3.2	Gage
103	Height of coils, max.		0.40%	3.2	Gage
104	Total height, max.		0.40%	3.2	Gage
Minor					
201	Poor workmanship		1.0%	3.8	Visual

REMARKS

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PAGE 37

4.4.5.1, line 7: Delete “The lot shall be rejected in one (1) or more ... ” and substitute: “The lot shall be rejected if two (2) or more ...”.

4.4.5.2, line 7: Delete “The lot shall be rejected if two (2) or more ...” and substitute “The lot shall be rejected if three (3) or more ...".

PAGE 39

4.5.2.1 Charge composition. Line 3, delete “Method 102.1” and substitute “Method 102.2.1”.

PAGE 40

4.5.2.6: Delete in its entirety.

PAGE 41

4.5.6, line 2: Delete "(70 ± 5°F)".

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